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# Constraints on the structure of the Azores spreading center from gravity data

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**Abstract**

The Azores Archipelago is believed to be the site of the third arm of a Triple Junction between the Eurasia/Africa/North America plates. However, to the present no study has been able to identify its segmentation pattern, the spreading mechanism and its relationship with the well-known topographic features of the Mid-Atlantic Ridge (MAR). Here we present a new gravity compilation obtained with the existing National Geophysical Data Center (NGDC) data, merged with new gravity profiles collected during the ESCAPE cruise in 1995. This compilation is used to calculate a Free Air Anomaly (FAA) map, which is used to test two different models, the Mantle Bouguer Anomaly model and the elastic plate model, for the study of the thermal regime of the Terceira Axis. The analysis of the results from both models demonstrates that the elastic plate model successfully models the gravity data from the Azores Plateau and that there is no gravity evidence for the existence of a spreading axis. The elastic plate thickness $T_e$, with a value of 7–8 km, suggests a very young lithosphere (about 10 Ma) at the time of the load of the Azores Plateau.

## Introduction

The FAA over the axial valley of the slow spreading ridges, as is the case of the Mid-Atlantic Ridge (MAR), are typically negative with amplitudes of several tenths of mGal and with peak to peak horizontal distances of approximately 50 km (Hall et al., 1986). The amplitude of these anomalies decrease for larger spreading rates, and in fast spreading ridges it reach positive values of the order of magnitude of 10 mGal.

The relationship between the ridge spreading rate and the shape of gravity anomalies is, in some way, a consequence of its relationship with the ridge's morphology. However, the morphology is not only a function of this parameter. It reflects also the local conditions of magma supply and so, where the thermal regime is abnormal, particularly near hot spot areas, the characteristics of the gravity anomalies are not a simple function of the spreading rate. It was shown (Lewis, 1981, 1982; Madsen et al., 1984; Hall et al., 1986) that the amplitude of free air anomalies can not be explained only in terms of topography. This implies that either some density contrast, other than the topography, contributes to the FAA, or that regional compensation models must be applied.

Information on the density structure of the ridge axis and on variations in crustal thickness is usually obtained by reduction of gravity data to the residual gravity anomaly. The residual gravity anomaly is obtained by removing three 3-D effects from the free air anomaly. The first two are the contributions from the water-crust and crust-mantle interfaces, assuming a constant crustal thickness and density, and what is obtained is the so-called Mantle Bouguer Anomaly (MBA). This correction represents the anomaly that would be measured if the layer between the sea surface and the Moho was filled with a material having the same density as the mantle (Prince and Forsyth, 1988). The third correction is the removal of the effect of lithospheric cooling with age. After these

three corrections the remaining signal, that means the residual gravity anomaly, is then interpreted in terms of along axis variations of crustal thickness and/or density anomalies (Kuo and Forsyth, 1988).

In the studies of the South Atlantic Ridge, Kuo and Forsyth (1988) were the first to conclude for the existence of a correlation between residual anomalies and ridge segmentation. They noted that the centers of the segments are typically associated with negative circular anomalies, and that the discontinuity zones and the off axis area are marked by positive anomalies. This gravity signature was interpreted as the result of crustal thinning along the axis, that can reach almost 50% over the discontinuities that bound the segments (Lin et al., 1990). However, the negative anomalies do not always seem to be centred over the segments, and sometimes they show some lateral offset in relation to the strike of the segments (e.g., Morris and Detrick, 1991; Detrick et al., 1995).

With a different approach, Cochran (1979), McNutt (1979), and Lewis (1982) have shown that the gravity anomalies observed over the East Pacific Rise could be explained by a regional model in which the topographic load of the ridge is supported locally by stresses within the plate. The best fit between the gravity anomalies and those deduced from the elastic plate model appear to require an equivalent elastic thickness of the order of 2–6 km. Similar calculations for the MAR (McKenzie and Bowin, 1976; Cochran, 1979; Kostoglodov et al., 1981) showed that elastic thickness between 7 and 10 km were necessary to achieve a good fit between the model and the observations.

There are, however, two objections against the elastic model as an explanation for the origin of the ridge axis gravity anomalies (Lewis, 1982; Madsen et al., 1984). First, if the existing thermal models are correct, then the mechanical strength of the plate near the ridge axis must be very low, and therefore it is unlikely that the plate rigidity is enough to support the axial load. Second, the elastic model predicts a crustal thickening towards the rise axis due to the deflection of the crust mantle boundary beneath the axial topography.

In its simplest form, these two approaches to model the observed FAA are different in the assumptions they make on the form of the crust mantle interface. The MBA model invokes a crust whose bottom boundary (the Moho) follows the sea-bottom topography, and the elastic plate model assumes that the bottom boundary is in some sort a mirror of the topography. The MBA can be seen as a way of processing gravity data in which the deviations from the initial assumptions are either interpreted as variations of crustal thickness and/or mantle density structure. However, many authors used this approach in a way similar to the elastic plate model with proven good results.

Due to its difficult physical interpretation and the fact that there is no seismic evidence for significant crustal thickening beneath the ridge axis, the use of the elastic plate model was hardly sustained when studies of isostatic compensation were carried out at Mid-Oceanic Ridges. Neumann and Forsyth (1993) synthesized this situation when they called it the ‘paradox of the axial profile’: while the quantitative relationship between along-axis topography and gravity can be explained by the flexural response of a thin, narrow elastic strip, representing the inner rift valley isolated from the rest of the plate, across-axis variations show that the median valley itself is not compensated. These authors presented a non linear, finite element, extensional model of the median valley formation that explains simultaneously the observed topographic and gravity variations of the MAR from 31° to 36° S when the dynamic mechanism creating the median valley topography is taken into account. Away from the ridge axis there is no paradox and the elastic plate model seems very realistic and has been widely used in studies of isostatic compensation of topographic features such as hotspot volcanoes.

In this study we examine the isostasy of the Azores Plateau, which contains the third arm the Azores Triple Junction and which is believed to be part of a Ridge-Ridge-Ridge (RRR) triple junction. The global kinematic models like the NUVEL1 (DeMets et al., 1990), imply for the motion in the Azores boundary, a right lateral transtensional regime along the Eurasia (EU)/African (AF) boundary, with an extensional component of 4 mm/yr full spreading rate (see Figure 1). The purpose of this paper is to examine the results of both compensation models, and analyze, from the merits of each one, the deep structure of the Terceira axis.

## Data

Figure 1 shows the location of the major tectonic features of the Azores Plateau, a sketch of the plate boundary between EU and AF plates. It also shows the direction of the relative plate motion as well as the zone selected for this study and the gravity data location.

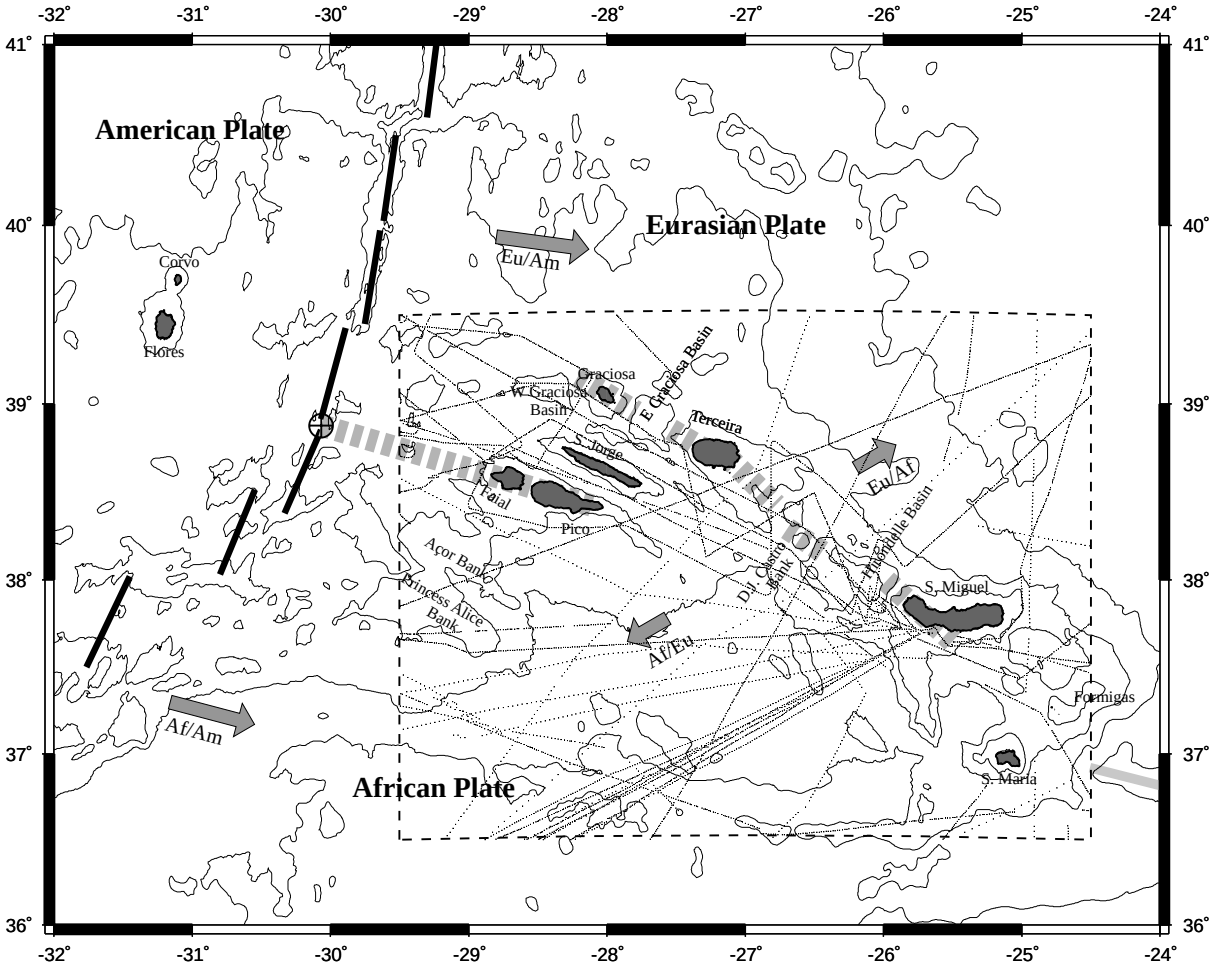


*Figure 1.* Regional setting of the Azores Plateau showing the location of the major tectonic features. The heavy lines represent the MAR. The arrows represent the velocity of relative plate motions. In order to be still visible Eu/Af arrow is exaggerated comparing to the Eu/Am arrow. The light grey thick line is a sketch of Eu/Af plate boundary. The doted lines inside the rectangle indicate the location of gravity profiles used to construct the map of Figure 5.

Surface ship gravity profiles were obtained from the National Geophysical Data Center (NGDC) or were collected in 1995 during ESCAPE mission of MARFLUX project (Bougault et al., 1996). Bathymetric data were obtained from the same sources, together with GEBCO charts and multibeam tracks collected during HEAT mission in 1994 (German et al., 1996). The region depicted in the inset of Figure 1 was selected to this study because there is enough data available and because we wanted to investigate the tectonic setting of the Azores Plateau away from the MAR axis.

Figure 2 shows the bathymetric map (Lourenço et al., 1999) of the survey area contoured at a 500 m interval. This area contains the EU/AF plate boundary, which Krause and Watkins (1970) interpreted as a spreading center trending WNW-ESE. The morphology of the Azores spreading center consists in a succession of basins and islands that were collectively called the Terceira Rift. Away from this topographic feature the bathymetric chart shows that the sea floor deepens rapidly in the NE–SW direction. However, SW of Faial and Pico islands the picture is confused by shallow banks which are part of a structure known as Faial Ridge that flanks the Mid-Atlantic Ridge from the Azores down to the FAMOUS area.

To compute the FAA map it is essential to reconcile the gravity data from different ship tracks. This is particularly needed in the present case, since the data set comes from several collecting agencies, from

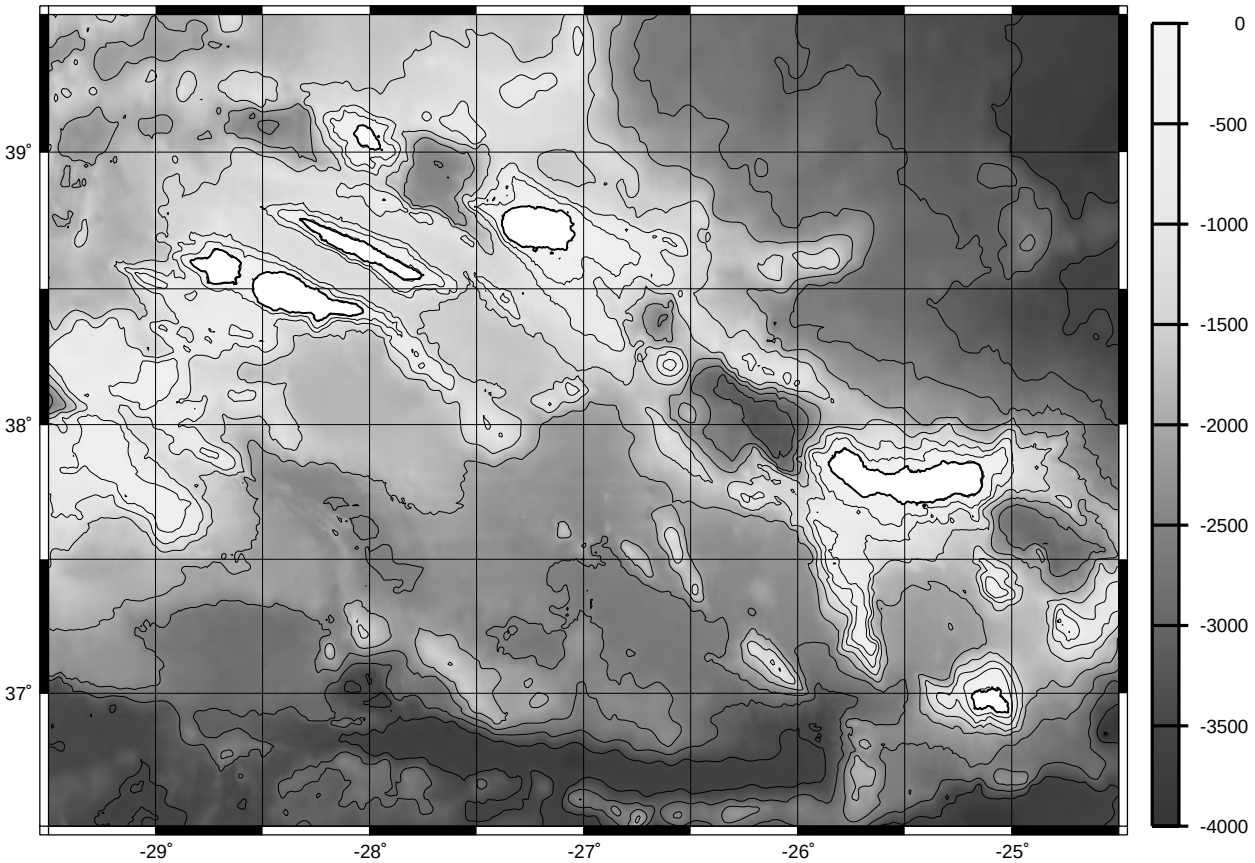


*Figure 2.* Bathymetry map contoured every 250 m. The map was constructed by interpolating the new compilation of Lourenço et al., (1999).

which the positioning, the gravimeters type and the calibrations made, vary significantly. To minimize the crossover errors we employed the method of Wessel (1989). The principle of this method is based on the assumption that a simple model having only two parameters can explain most of crossover errors. These are a constant shift and a drift-rate that apply for the duration of the cruise. These parameters are determined for each leg in such a way to minimize the crossover errors in a least squares sense.

Figure 3 shows the histograms of the crossing errors for both internal and external crossovers before and after correction. The standard deviation error was 13.7 mGal before correction, and after applying the minimization procedure the crossing error was reduced to 6.9 mGal.

The FAA map shown in Figure 5 was produced by interpolation of the corrected gravity onto a 1.8-km grid using Smith and Wessel (1990) gridding algorithm, and contoured at 10 mGal interval. As expected, the effects of the short-wavelength features in the bathymetry dominate the free-air anomaly. The highest values correlate with the archipelago axis and the sea mounts, and the lowest values correlate with the basins and with the trace of the East Azores Fracture Zone.

We decided not to include in the compilation any satellite-derived gravity data. Sandwell and Smith (1997), while discussing the accuracy and resolution of their gravity grid version 7.2, compared it with several shipboard gravity profiles. They found that the individual ship profiles show rms differences of 3–6 mGal depending on the proximity to a stacked altimeter profile. In the ESCAPE cruise mentioned before, we followed the trackline of two ERS-1 profiles for a distance of approximately 80 km. The comparison between the two gravity data sets is shown in Figure 4. The rms difference is in the range found by Sandwell and Smith (4.9 mGal) for the west track but it is rather higher for the east track (11.4 mGal). These differences, however, are not very significant because they are computed for shorter distances than

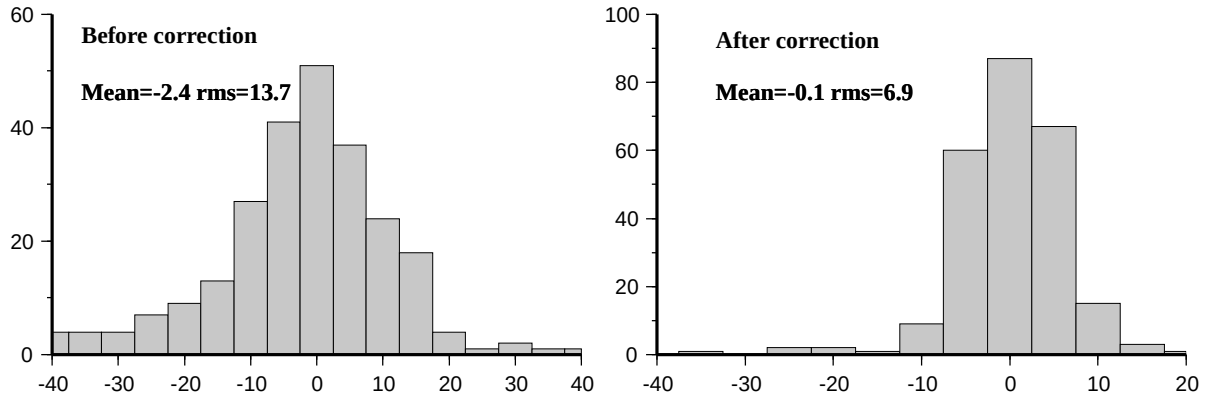


*Figure 3.* Histogram of cross-over errors of free-air anomaly before and after correction. The respective rms misfits are written in each diagram.

the ones used by Sandwell and Smith and because they correspond to a zone where the bathimetry has a strong content in the short wavelength domain, that is absent in the satellite-derived gravity field.

The reason why we did not use any satellite-derived gravity data is because the main purpose of this work is to study the mechanism of the isostatic compensation in the Azores plateau, and not to establish a detailed gravity map of the area. To carry out this study it is very important to know the bathymetry in order to compute the gravity anomaly due to a crustal model. Using satellite gravity grid data implies the use of a bathymetric grid for which there is no guaranty that for each gravity node the corresponding bathymetric node is constrained by a 'real' data point. Obviously this happens only when the bathymetry available is not derived from multibeam data, as is the case of the grid we used. In fact this is the same problem that will be addressed in the next section when discussing the errors accumulated in the grid to grid operations away from ship tracks.

## Isostatic compensation in the Azores Plateau

The first step to study the isostatic compensation in the Azores Plateau is the computation of the mantle Bouguer anomaly. To do so, we used the bathymetric grid to calculate the gravitational attractions of the water-crust and crust-mantle interfaces using Parker's (1972) method, retaining up to the forth non-linear term. The crust was assumed to have a constant thickness of 6 km and a constant density of 2700 kg m$^{-3}$. Assumed densities for the sea water and mantle are respectively 1030 kg m$^{-3}$ and 3300 kg m$^{-3}$. The effects of the two interfaces were then subtracted from the FAA grid to generate the mantle Bouguer anomalies.

However, this easy and fast grid to grid operation accumulates errors away from the ship tracks (Kuo and Forsyth, 1988). This arises because in every step the topographic grid nodes, which are artificially determined by the interpolation algorithm away from the data points, will contribute with a signal that will be subtracted to a FAA also obtained only by interpolation. In order to clean these artifacts, we interpolated the MBA grid along the original profiles (where the FAA was actually measured), and we computed a final MBA map based on these data only (Figure 5, bottom). Finally, the mean grid value was removed, as only relative variations are meaningful.

The dominant feature of the MBA is the strong correlation with the topography. Prominent lows are associated with the archipelago axis, and highs are located on both flanks of the Plateau where the bathymetry is deepest. Also, two large wavelength components are present in the map: the first one trends perpendicularly to the archipelago axis and the second, more discrete, is perpendicular to the MAR. Such long wavelengths are normally attributed to the cooling and thickening of the lithosphere with increasing age of the seafloor. In the present case, however, this picture is complicated by the fact that we may have two close ridges – the MAR and the Terceira spreading center – away of which one the lithosphere is supposed to get thicker.

In order to estimate the effect resulting from the depth variations of the lithosphere-astenosphere boundary, we followed the method of Rommevaux et al. (1994). In this method the depth of the boundary is given by the well known expression $y_L = 2.32\sqrt{kt}$ (Turcotte & Schubert, 1982), where $t$ is the age of

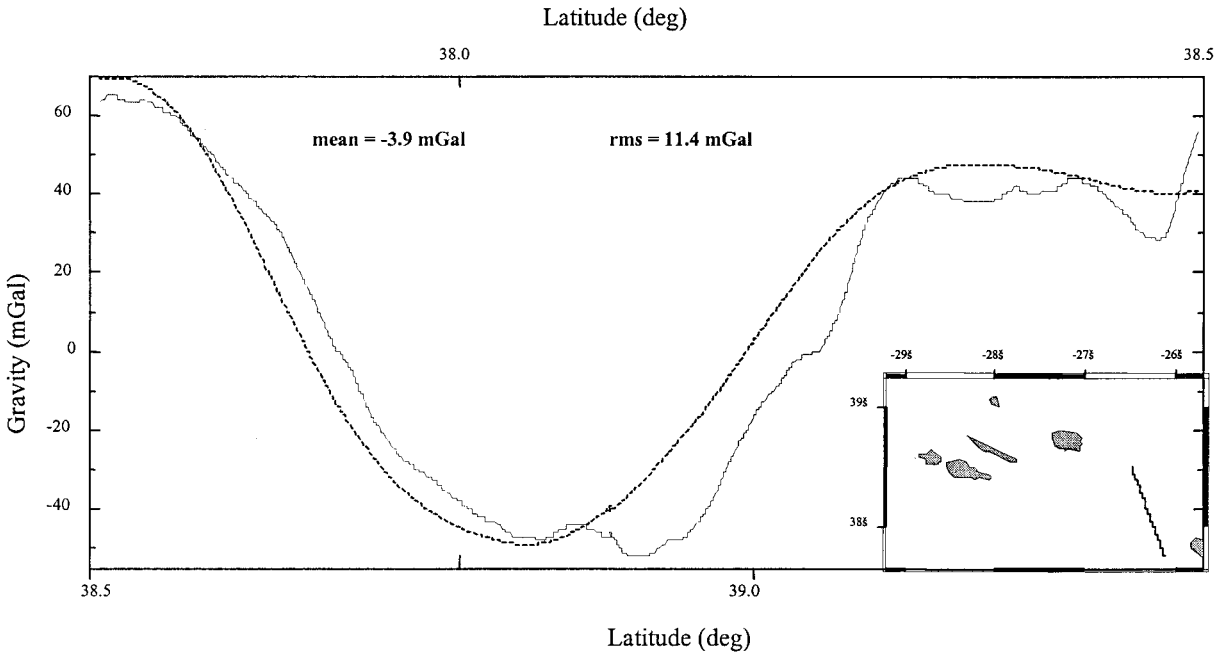


*Figure 4.* Comparison between a shipboard gravity profile (solid line) and a coincidentally located satellite-derived gravity (dashed).

the lithosphere (expressed in Ma), $k$ is the thermal diffusivity (assumed to be $0.8 \times 10^{-2}$ cm$^2$ s$^{-1}$). The contrast of density at this interface was assumed to be $-60$ kg m$^{-3}$.

Magnetic anomaly identification from anomaly 2A to anomaly 20 over the Azores Plateau (Luis, 1996) allows us to generate a first lithospheric age grid in which the effect of the Terceira ridge was not taken into account.

The main question now is how to model the cooling effect perpendicular to the Terceira ridge because there are no continuous isochron anomalies parallel to this axis. The magnetic anomaly map (Luis, 1996), however, reveals that away from the topographic highs some magnetic isochrons can be followed across the Azores Plateau. This means that only a narrow band surrounding Terceira axis, and distributed around the main volcanic structures, can be considered as young crust. So we made the assumption that this band has zero age and that the age of the region outside is determined by the age of the magnetic isochrons. This is obviously a crude assumption but it places an upper bound to the cooling effect.

The gravity signal resulting from the combined effect of the MAR and the Terceira axis on the depth of the lithosphere-asthenosphere boundary is shown in Figure 6. This theoretical gravity field exhibits age dependency with wavelengths similar to those represented in Figure 5 (bottom). However, as can be seen in the map of the residual MBA (Figure 7, top), there is still a strong wavelength content in the residual field that is orthogonal to the Terceira Axis. This means that the cooling effect can not be the only responsible for the long wavelength gravity residuals and therefore a different mechanism must be considered.

A different approach can be made by the use of the elastic plate model. Here the lithosphere is considered to be made of two layers, an upper layer which is able to sustain elastic strain at time scales of the deformation due to a topographic load, and a lower layer where the strains are released at time scales shorter than that of the deformation process. The formalism that is needed to represent this model is rather complex (McKenzie & Bowin, 1976; Watts, 1978) but it can be simplified if we consider the case of thin plates (Banks et al., 1977), that means, a plate which thickness is small when compared to the deformation wavelength. The two models are different concerning the amplitude of the deformation in the small wavelength range. However, the upward continuation made in the computation of the gravity anomalies at sea level, minimizes the difference between both models (Lewis, 1982).

The most useful parameter to describe the mechanical behavior of the plate is the flexural rigidity $D$,

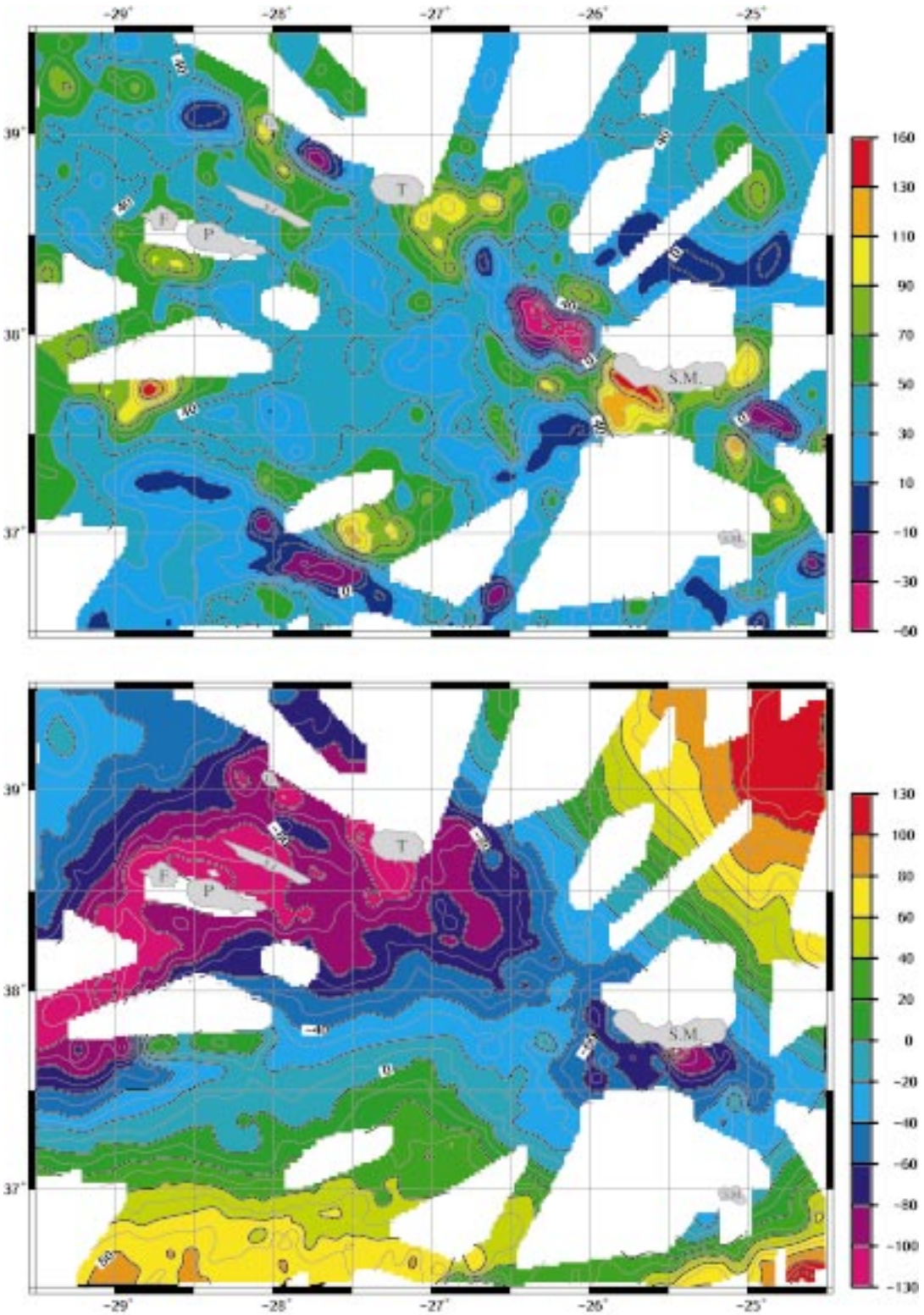


*Figure 5.* (top) Free-air gravity anomaly map (contour interval is 10 mGal. (bottom) Mantle Bouguer anomaly map contoured every 10 mGal. The zero level is arbitrary and corresponds to the mean value of the grid.

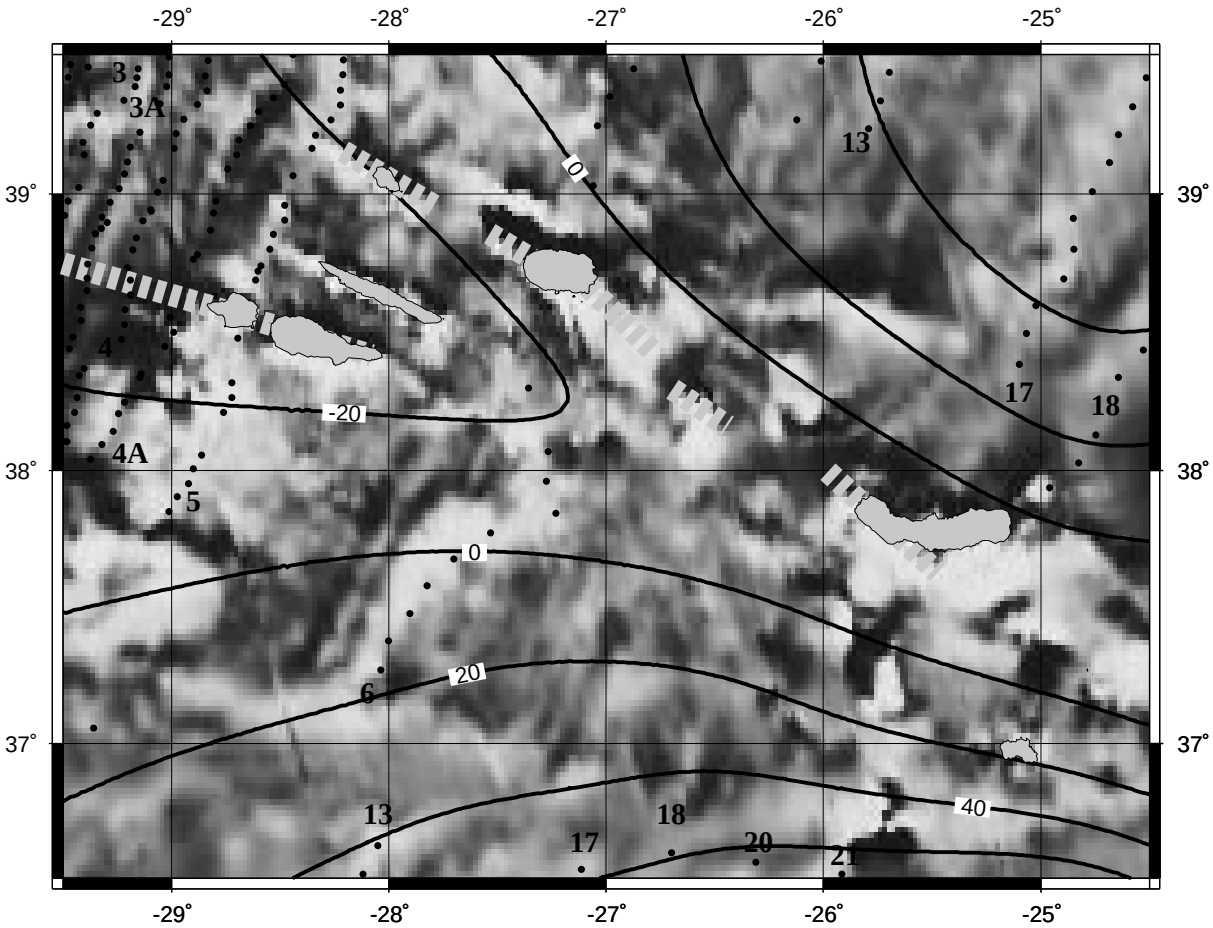


*Figure 6.* Predicted gravity anomaly due to lithospheric cooling and thickening with age away from both the MAR and the Terceira Axis. The background depicts the magnetic anomaly of Luis (1996) represented with an equalised grey scale. Lighter tones correspond to positive anomalies and darker tones to negative anomalies. The closed symbols locate the magnetic anomaly picks (isochrons).

which is a measure of the resistance of the lithosphere to deformation:

$$D = \frac{ET_e^3}{12(1-\nu^2)}, \tag{1}$$

where $E$ is the Young's modulus, $\nu$ is the Poisson's ratio and $T_e$ the equivalent elastic thickness.

In the Fourier domain the deflection $W$ of a thin elastic plate under the effect of the seafloor load is related with the bathymetry by the equation:

$$W(\vec{k} = \frac{-g(\rho_c - \rho_w)}{D|\vec{k}|^4 + g(\rho_m - \rho_c)} B(\vec{k}), \tag{2}$$

where $k$ is the modulus of the horizontal wave vector; $\rho_w$, $\rho_c$, $\rho_m$ are, respectively, the densities of the water, crust and mantle, $g$ is gravity, and $B(k)$ is the two-dimensional Fourier Transform of the bathymetry. The gravity effect generated at the surface by the interface located at the mean depth $d$ which form is described by $w(x, y)$, may be determined using Parker's method (1972):

$$F(g(x, y)) = 2\pi G \Delta\rho e^{-|\vec{k}|d} \sum_{n=1}^{\infty} \frac{|\vec{k}|^{n-1}}{n!} F(w(x, y)), \tag{3}$$

where $\Delta\rho$ is the density contrast and G is the gravity constant.

Three parameters are relevant to calculate the gravity anomaly due to the plate deflection: the mean thickness of the crust $d$ (compensation depth), the density contrast $\Delta\rho$ at the crust mantle interface and the equivalent elastic thickness $T_e$. We allowed $T_e$, the Moho depth $d$ and the density of the crust, $\rho_c$, to range from 0 to 20 km and 6 to 22 km and 2500 to

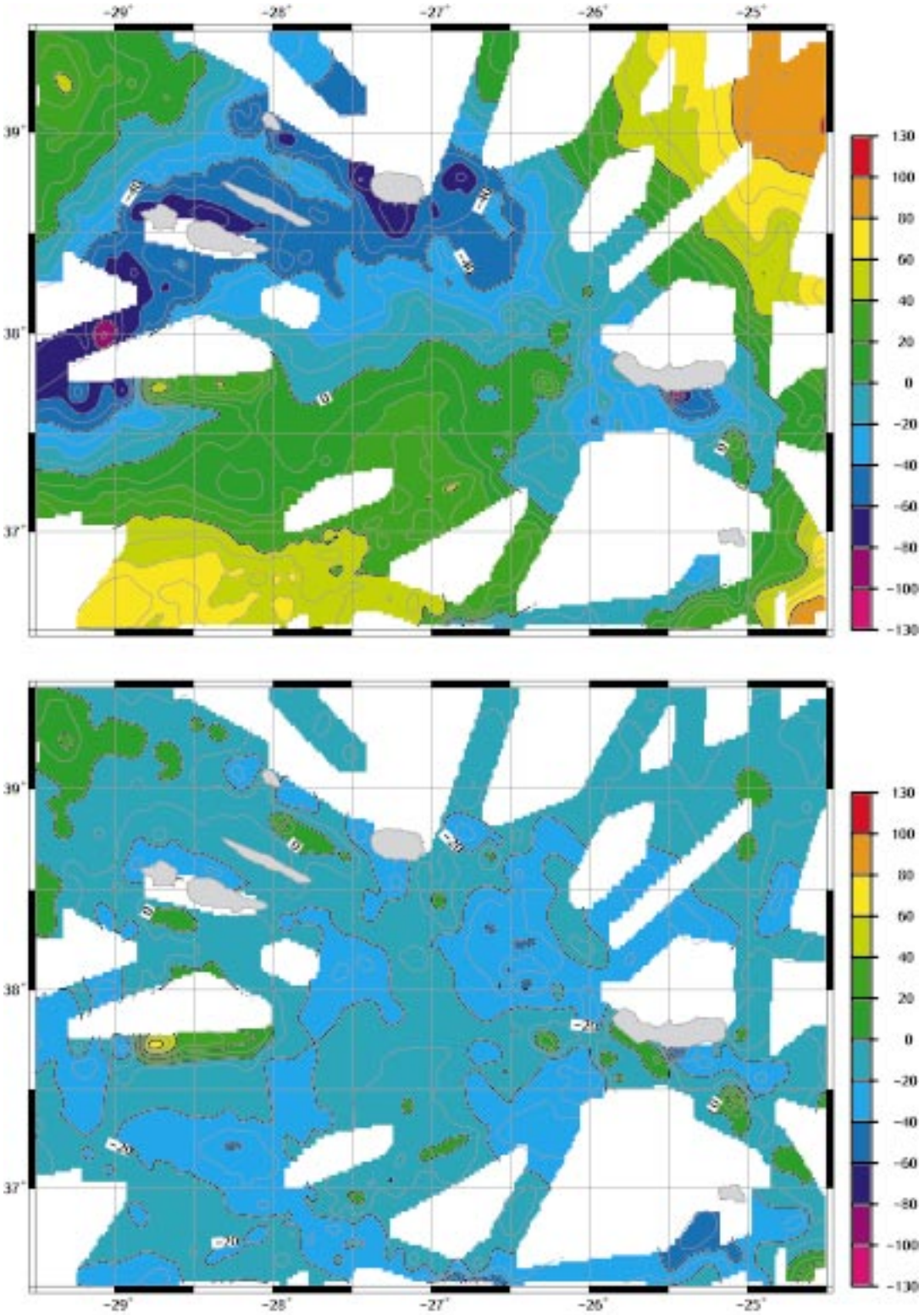


*Figure 7.* (top) Residual MBA (contour interval is 10 mGal). This residual anomaly is obtained by subtracting the MBA of Figure 5 from the gravity effect of the crustal thickness variation displayed in Figure 6. The zero level is arbitrary. (bottom) Residual anomaly for the elastic plate model (contour interval is 10 mGal). This residual anomaly is obtained by subtracting the observed free-air anomaly from the theoretical anomaly of the elastic plate model ($T_e = 7$ km; $d = 12$ km; $\rho_c = 2700$ kg m$^{-3}$; $\rho_m = 3300$ kg m$^{-3}$). The zero level is arbitrary.

2700 kg m$^{-3}$, respectively. After calculating the theoretical gravity anomalies due to the effects of the sea floor bathymetry and the plate deflection, we interpolated the grid points onto the ship tracks and subtracted them from the free air anomaly.

The theoretical model best fitting the gravity anomaly was determined using the RMS measure of the misfit (Figure 8). All of the misfit plots show a well defined minimum which is obtained for a Moho depth of 12–14 km. Considering that the average depth to the topography is about 2 km, this leads to an estimation of 10–12 km for the mean thickness of the crust. The best fitting elastic plate thickness value changes very little with $\rho_c$ and is constrained between 7 and 8 km. Based on the misfit curves, a model with $\rho_c = 2700$ kg m$^{-3}$, $d = 14$ km and $T_e = 7$ km provides an excellent fit to the data. The residual gravity anomaly obtained with this model is shown in Figure 7 (bottom).

## Discussion and conclusions

To model the FAA we have first assumed a crust with a constant thickness and that the lateral density variations in the crust and mantle were a function solely of the temperature variations, which depend only on the age of the lithosphere. The residual anomalies obtained with this model (Figure 7, top), that reflect the departures from these assumptions, are very high and still contain a strong wavelength component perpendicular to the Terceira axis. In fact, along the NE–SW direction and over a distance of about 500 km, we can see that the residual anomaly reaches 100 mGal at the NE corner of the area, decreases to values around −40 mGal at the central part, and then increases again to 80 mGal at the SW corner of the map.

The residual anomaly is traditionally interpreted in terms of crustal thickness variation. This interpretation, however, would imply accepting abnormally high values for the crustal thickness and, mostly, an unreasonably width rise axis for the Azores spreading center.

These comments are obviously appropriate if we consider that the Azores spreading center behaves, in some way, like a ‘normal’ spreading ridge. However, the real nature of the structure of the Terceira Rift is still a matter of debate. Global plate kinematics considerations argue for the hypothesis of the Terceira Rift being a very slow and oblique spreading ridge, acting as the third arm of the Azores Triple Junction. However, existing magnetic and seismic data cast many doubts on this simple interpretation.

The magnetic anomalies over the Terceira Ridge reflect the en-echelon pattern of the morphology (Luis, 1996). Sharp positive anomalies are centered over the top of the highs, but there is, definitely, no pattern of Vine and Mathews anomalies parallel to the Terceira Ridge. However, the Azores spreading center has affected the lithosphere along a narrow band trending WNW-ESE, not much wider than the rhombic area delimited by the emplacement of the islands. Outside this band, namely to the north-east, the magnetic isochrons are easy to identify. For example, in the case of anomaly 18, it comes to end very close to S. Miguel island (Figure 6). This means that the accretionary processes related to the Azores spreading center activity are really confined to a narrow area around the morphological highs.

The seismotectonic studies of this region reinforce this conclusion: they show a distribution of earthquakes with epicenters concentrated in a narrow zone aligned with the islands. The focal mechanisms, however, show a great diversity, with solutions ranging between WNW–ESE normal faulting, E–W right-lateral strike-slip (Udias et al., 1976; Grimison and Chen, 1986), and NNW-SSE left-lateral strike-slip motion (Hirn et al., 1980). For the first group of focal mechanisms the direction of the tensional axes is compatible with the general direction of the relative motion between Eurasia and Africa, but that is no longer the case for the second kind of mechanisms (Figure 9).

Miranda et al. (1999) have proposed a model for the present functioning of the Azores spreading center based on data from a 34-day long Ocean Bottom Seismometer array deployment. This model concludes that the tectonic setting of the spreading center is characterized by the existence of two set of faults, in the N120E and N150E directions, which define a series of tectonic blocks who relative motion accommodates the interaction of the Eurasian and African plates. The build up of the structural features that composes the Terceira Rift is thus tectonically controlled and there is no evidence of a ‘normal’ segmentation pattern neither on the morphology nor on the magnetic anomalies.

The dominant feature of the residual MBA is its extremely large amplitudes (both positive and negative) and its long wavelength. According to Kuo & Forsyth (1988) and Rommevaux et al. (1994) the errors that can be generated by the method followed on the computation of the gravity anomaly are of the order of magnitude of 7–8 mGal. As the obtained values for the

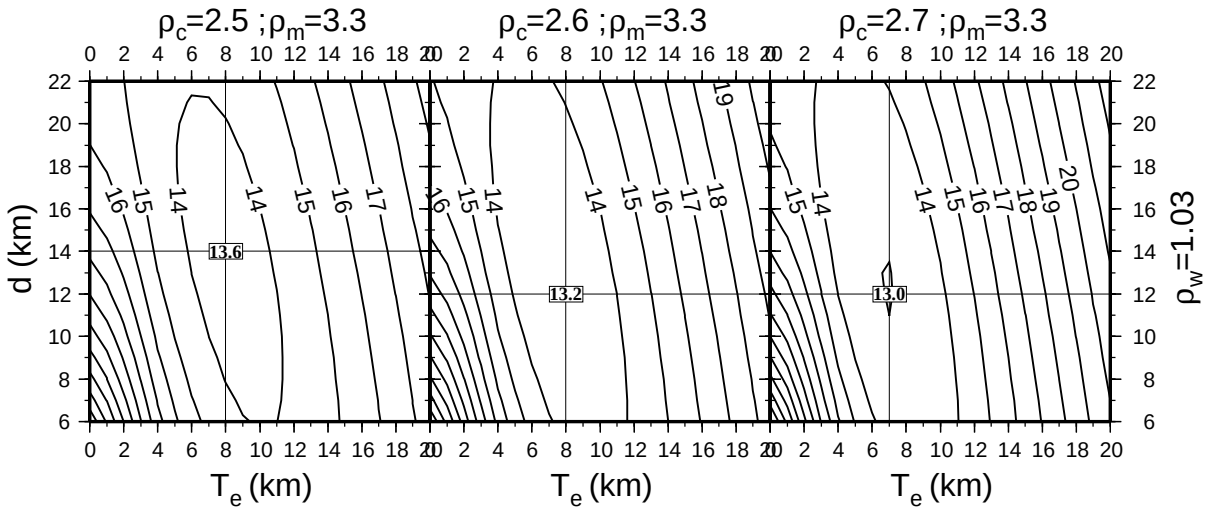


*Figure 8.* Root-mean-square misfit in mGal of the elastic plate model as a function of the assumed elastic plate thickness $T_e$, the depth of compensation $d$ and density of crust $\rho_c$ and mantle $\rho_m$.

residuals are well above the accuracy estimated, we can conclude that they are rather large and so any interpretation model for the area must be able to explain these values.

The large amplitude of the residual MBA, inversely correlated with the long wavelength bathymetry, emphasizes that either the assumed thermal model is not appropriate or that the overall structure of the Azores Plateau is very distinct from the assumptions made in the calculation of the residual MBA. We favor the second explanation because, even if the fine structure of the age distribution in the immediate vicinity of the islands is not well know, we do have a strong constraint on the age of the crust some kilometers away from the topographic highs. So, we can conclude that the cooling effect generated by the Terceira rift can not justify the amplitudes of the MBA residuals.

On the contrary, we successfully modeled the gravity data from the Azores plateau with the elastic plate model. Here the Azores Plateau, as a whole, does not present any significant residual anomaly. Only in the small wave number content we can find significant positive values over the ridge south of Pico-Faial islands, and strong and negative anomalies over the basins between the islands and along the East Azores Fracture Zone. These anomalies must reflect the lack of local isostatic compensation as expected.

One aspect crucial to the determination of the hypocenters of the seismic activity recorded during the OBS deployment, mentioned above, was the choice of the velocity model. It was found that the available models for the zone (Hirn et al., 1980; Purdy and Detrick, 1986) gave hypocenters depths to large and very dependent upon the model used. It was then necessary to calculate a new velocity model. In this model the crust-mantel interface, or Moho, is located at a mean depth of 11 km below sea-bottom. This figure is in a very good agreement with the depth of that interface as determined from the elastic plate model (Figure 8). Remember that the best value found, from the RMS misfit, was 12–14 km but from this figure we have to subtract the mean sea depth (about 2 km), which leaves us with a mean crust thickness of about 10–12 km. This value, however, is larger than the 8 km found by Searle (1976) from a Rayleigh-Wave dispersion study.

These aspects reinforce the conclusion that the Terceira Axis mainly constitutes a zone of intense volcanism that partially absorbs the differential movement between the Eurasian and the African plates. This is readily confirmed by the analysis of the focal mechanisms displayed in Figure 9 where we can see that the section between S. Miguel and Terceira islands shows solutions compatibles with relative plate motion but where the other focal mechanisms, either to West or East of that section, exhibit a considerable range of solutions. Taking the zone defined by the seismicity in Figure 9 as the one containing the plate boundary and comparing the residual anomaly inside and outside it, we are forced to conclude that there is no gravity evidence for the existence of a spreading axis.

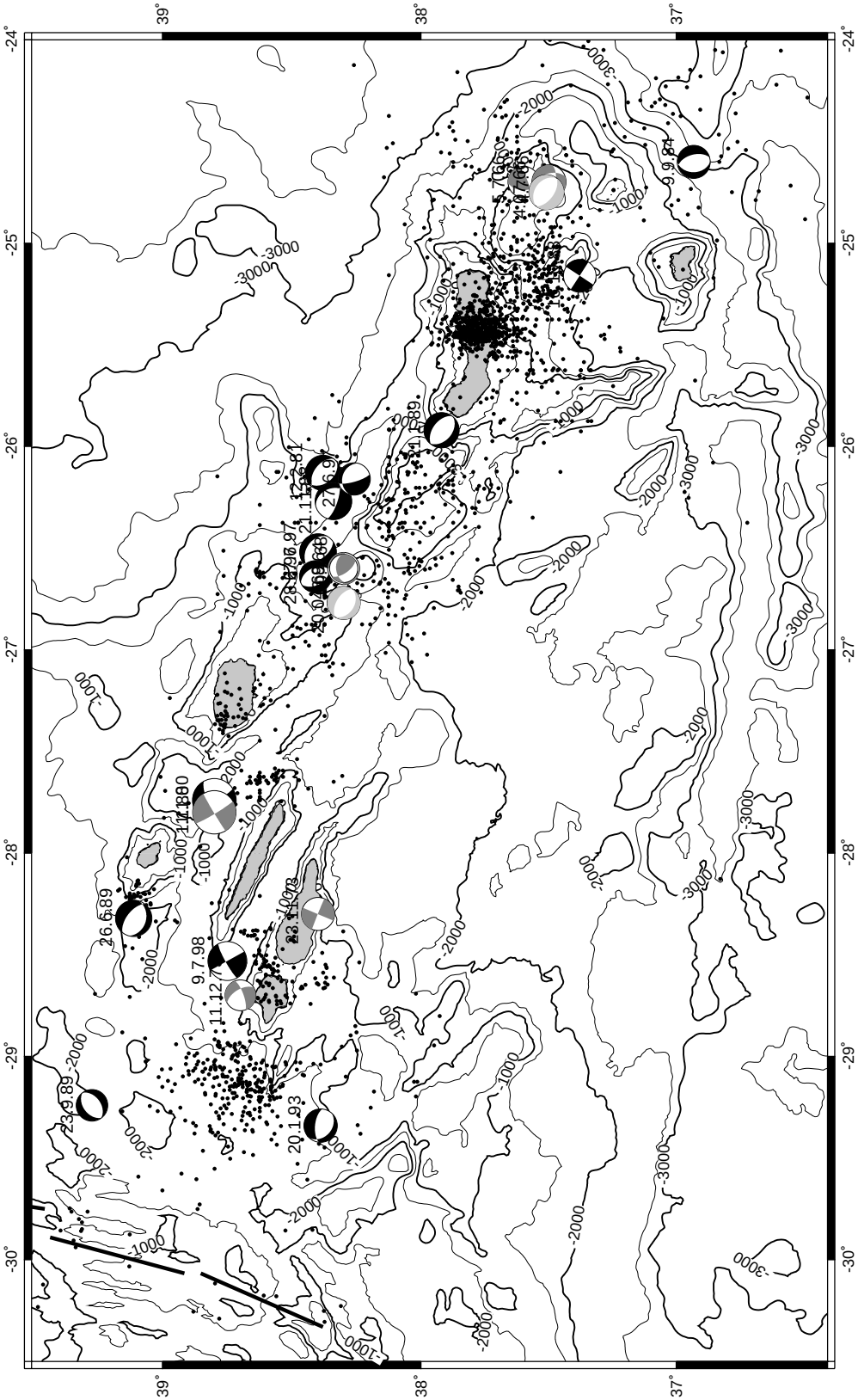


*Figure 9.* Seismicity map (M > 3) of the Azores plateau from 1980 until 1993 from the Azores seismic network. The focal mechanisms are extracted from the Harvard CMT solutions (black symbols), from Buforn et al. (1988) (dark grey) and from Grimison and Chen (1986) (light grey).

If we compare these conclusions with the analysis made by Géli et al. (1994) for the Mohns Ridge we can detect some similarities in both situations: there is an en-échelon pattern for both the topography and the magnetic anomalies and the active volcanic zones correspond to the topographic highs. Even the rhombic shape deep basins are found in both situations. Unfortunately these authors only calculated a residual map for the simple Bouguer anomaly so we cannot compare it with the results obtained in this study.

According to the Géli et al. (1994) the Mohns Ridge has a half-spreading rate of 7–8 mm $yr^{-1}$ and is considered one of the slowest along the mid-ocean ridge system. The NUVEL1 (DeMets et al., 1990) model applied to the Azores Spreading Center gives only a half spreading rate of 2 mm $yr^{-1}$. In this sense, we can think of the Terceira Rift as the limiting case of a ultra-slow spreading regime, where the accrectionary process are mainly controlled by regional tectonics. The diffuse character of the EU/AF plate boundary in this area is a reflex of the present complex stress pattern associated with the inherited structures from the past kinematic history. These factors are probably inhibiting the development of a typical mid-ocean ridge geometry. The topographic loads that result from that volcanism are than regionally compensated by a lithosphere made at the MAR and the overall gravity signature is better described by the elastic plate model.

Finally, as shown by Calmant and Cazenave (1987) the thickness of the elastic plate $T_e$ in kilometers supporting hot spot volcanoes increases roughly as $2.7 \pm 1.5\sqrt{t}$, where $t$ is the age in Ma of the lithosphere at the time of the load. Following this empirical relationship, the elastic thickness of 7–8 km found in this study indicate that the time of the Azores Plateau loading should have been between 7 and 9 Ma. This implies a rather young age to the Azores hot spot but it is in excellent agreement with the fact that the magnetic anomalies with an 'Azores trend' only disturb the magnetic isochrons up to anomaly 5 (Searle, 1980; Luis, 1996), which has an age of 10 Ma.

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